

Mock Final Exam 2 (Solutions)

(5:30 PM – 8:00 PM : 150 Minutes)

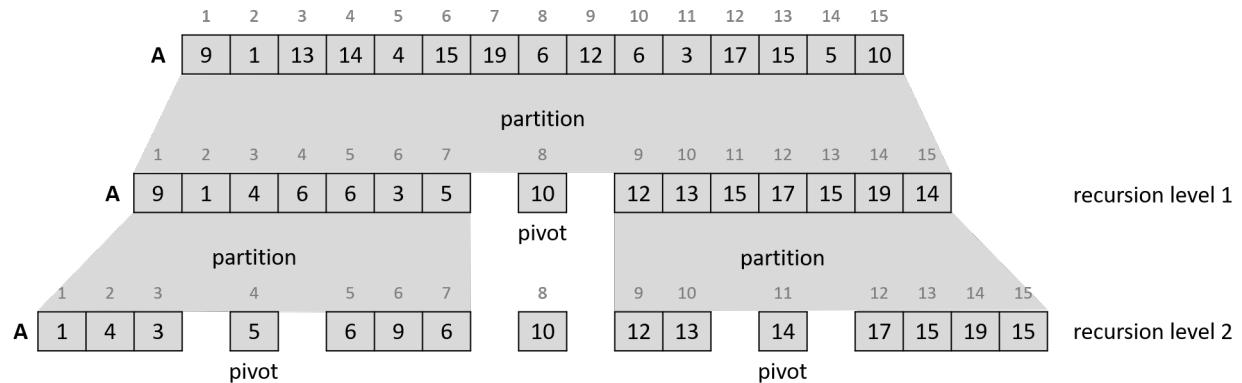
- This exam will account for 35% of your overall grade.
- There are six (6) questions, worth 250 points in total. Please answer all of them.
- This is a *closed book, closed notes* exam. *No cheat sheets* are allowed.
- You are allowed to *use scratch papers* for your calculations.
- You are *not allowed to use your own calculator*. A scientific calculator will be available inside the Respondus Lockdown Browser.

GOOD LUCK!

Question	Parts	Points
1. Partitioning	–	$2 \times 10 = 20$
2. Heapsort	$(a)-(j)$	$4 \times 10 = 40$
3. Prim's MST	$(a)-(l)$	$5 + 5 \times 10 + 10 = 65$
4. Topological Sort	$(a)-(k)$	$5 \times 10 + 5 = 55$
5. Matrix Chain Multiplication	$(a)-(c)$	$20 + 8 + 12 = 40$
6. Complexities	$(a)-(d)$	$7.5 \times 4 = 30$
Total		250

QUESTION 1. [20 Points] Partitioning. Recall the in-place partitioning algorithm we learnt in the class (before midterm) which is used by quicksort to partition the numbers in the array or subarray it is working on around the pivot. We always chose the last element of the given array/subarray as the pivot.

Given an array $A[1 : 15]$ as an input the following example shows the state of A after partitioning in the first two levels of recursion of quicksort.



You can describe the state of A after partitioning in recursion level 1 as:

[9, 1, 4, 6, 6, 3, 5] 10 [12, 13, 15, 17, 15, 19, 14]

Then after partitioning in recursion level 2 as:

[[1, 4, 3] 5 [6, 9, 6]] 10 [[12, 13] 14 [17, 15, 19, 15]]

Now given the following array $A[1 : 14]$ of numbers describe its state after partitioning in recursion levels 1 and 2.

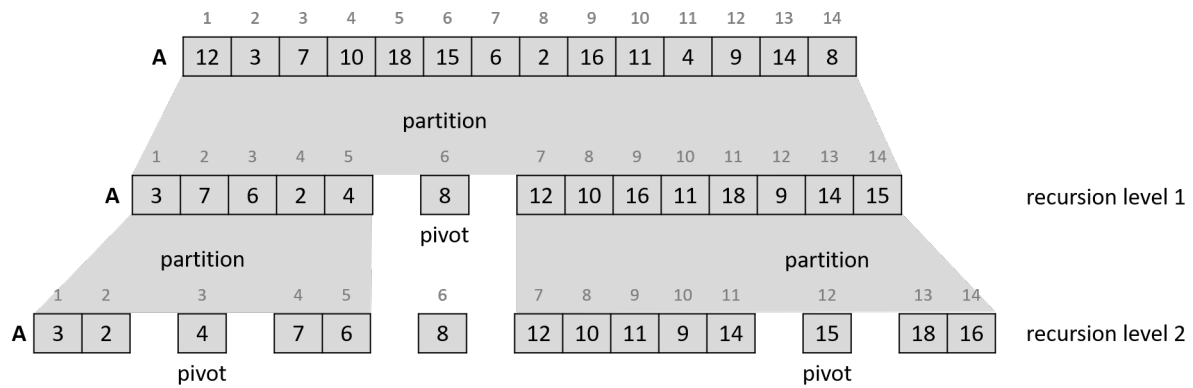
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	12	3	7	10	18	15	6	2	16	11	4	9	14	8

SOLUTION.

Recursion level 1: [3, 7, 6, 2, 4] 8 [12, 10, 16, 11, 18, 9, 14, 15]

Recursion level 2: [[3, 2] 4 [7, 6]] 8 [[12, 10, 11, 9, 14] 15 [18, 16]]

The solution above corresponds to the following diagram. You do not need to include this diagram in your answer.



QUESTION 2. [40 Points] Heapsort. Recall the heapsort algorithm we saw in the class. Given an array $A[1 : n]$ of n numbers as input first it turns it into a max-heap by calling BUILD-MAX-HEAP and then in each of the next $n - 1$ steps it removes the current maximum of the heap and heapifies the rest of the heap by calling MAX-HEAPIFY.

Now given the following array $A[1 : 10]$ as the input to the heapsort algorithm you are required to show the state of A after various steps of the algorithm.

	1	2	3	4	5	6	7	8	9	10
$A[1:10]:$	8	16	9	4	12	14	21	10	18	15

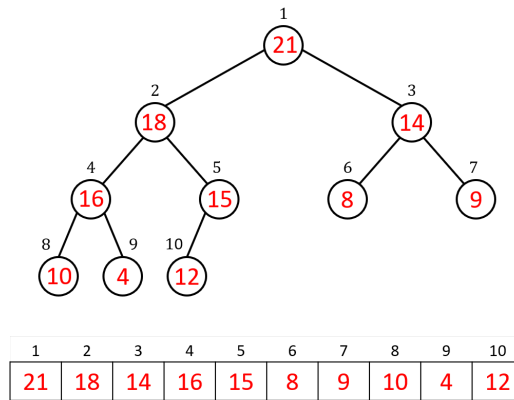
- 2(a) [4 Points] Show state of $A[1 : 10]$ after BUILD-MAX-HEAP.
- 2(b) [4 Points] Show state of $A[1 : 10]$ after removing the largest item of A from the heap and max-heapifying.
- 2(c) [4 Points] Show state of $A[1 : 10]$ after removing the 2nd largest item of A from the heap and max-heapifying.
- 2(d) [4 Points] Show state of $A[1 : 10]$ after removing the 3rd largest item of A from the heap and max-heapifying.
- 2(e) [4 Points] Show state of $A[1 : 10]$ after removing the 4th largest item of A from the heap and max-heapifying.
- 2(f) [4 Points] Show state of $A[1 : 10]$ after removing the 5th largest item of A from the heap and max-heapifying.
- 2(g) [4 Points] Show state of $A[1 : 10]$ after removing the 6th largest item of A from the heap and max-heapifying.
- 2(h) [4 Points] Show state of $A[1 : 10]$ after removing the 7th largest item of A from the heap and max-heapifying.
- 2(i) [4 Points] Show state of $A[1 : 10]$ after removing the 8th largest item of A from the heap and max-heapifying.
- 2(j) [4 Points] Show state of $A[1 : 10]$ after removing the 9th largest item of A from the heap and max-heapifying.

SOLUTION.

2(a) [4 Points] Show state of $A[1 : 10]$ after BUILD-MAX-HEAP.

21, 18, 14, 16, 15, 8, 9, 10, 4, 12

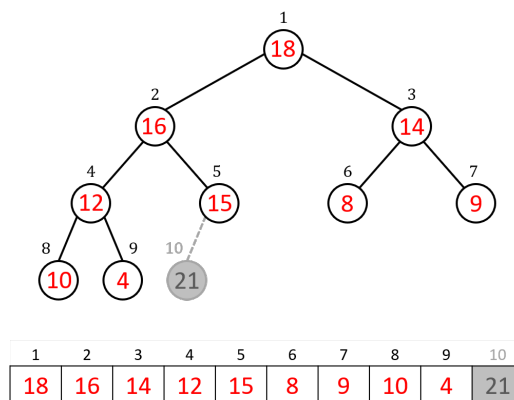
Diagram (no need to include in your answer):



2(b) [4 Points] Show state of $A[1 : 10]$ after removing the largest item of A from the heap and max-heapifying.

18, 16, 14, 12, 15, 8, 9, 10, 4, 21

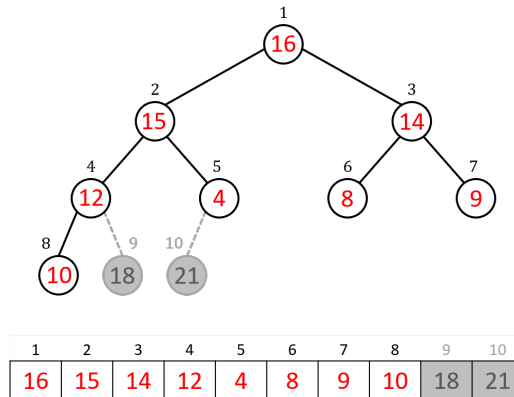
Diagram (no need to include in your answer):



2(c) [4 Points] Show state of $A[1 : 10]$ after removing the 2nd largest item of A from the heap and max-heapifying.

16, 15, 14, 12, 4, 8, 9, 10, 18, 21

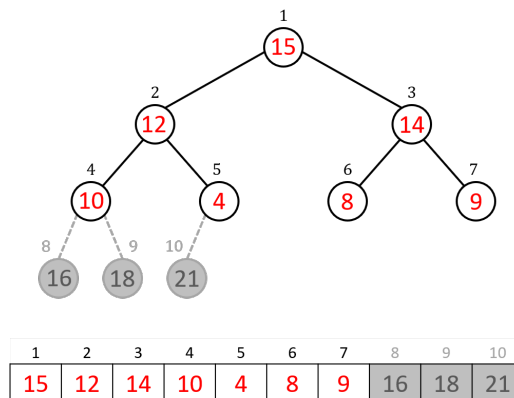
Diagram (no need to include in your answer):



2(d) [4 Points] Show state of $A[1 : 10]$ after removing the 3rd largest item of A from the heap and max-heapifying.

15, 12, 14, 10, 4, 8, 9, 16, 18, 21

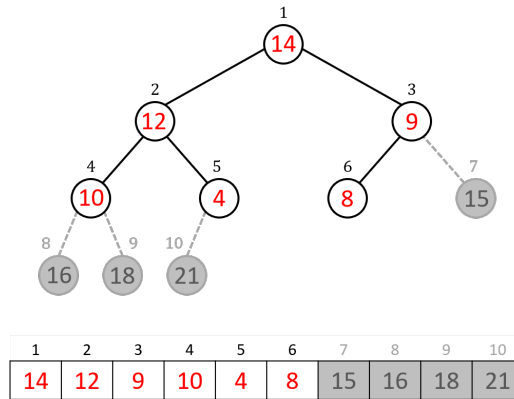
Diagram (no need to include in your answer):



2(e) [4 Points] Show state of $A[1 : 10]$ after removing the 4th largest item of A from the heap and max-heapifying.

14, 12, 9, 10, 4, 8, 15, 16, 18, 21

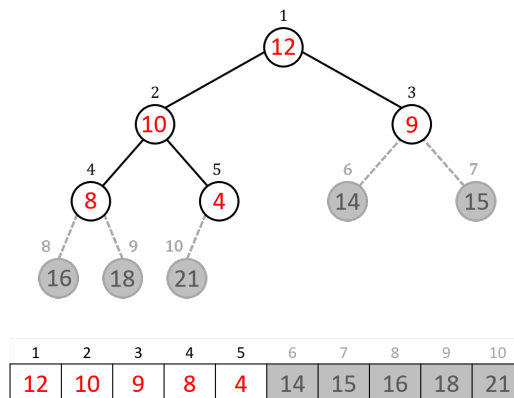
Diagram (no need to include in your answer):



2(f) [4 Points] Show state of $A[1 : 10]$ after removing the 5th largest item of A from the heap and max-heapifying.

12, 10, 9, 8, 4, 14, 15, 16, 18, 21

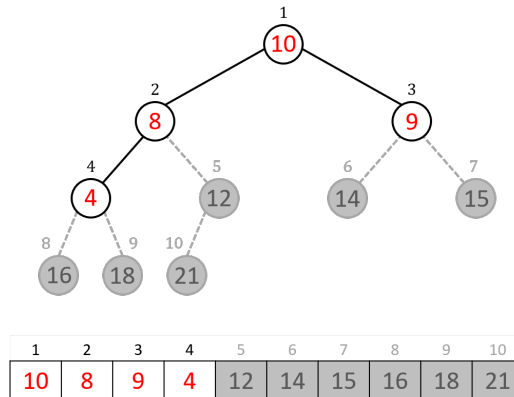
Diagram (no need to include in your answer):



2(g) [4 Points] Show state of $A[1 : 10]$ after removing the 6th largest item of A from the heap and max-heapifying.

10, 8, 9, 4, 12, 14, 15, 16, 18, 21

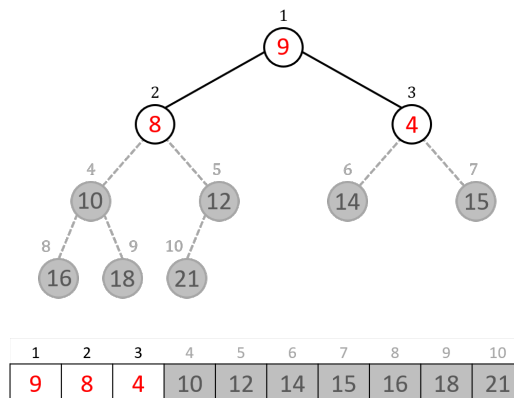
Diagram (no need to include in your answer):



2(h) [4 Points] Show state of $A[1 : 10]$ after removing the 7th largest item of A from the heap and max-heapifying.

9, 8, 4, 10, 12, 14, 15, 16, 18, 21

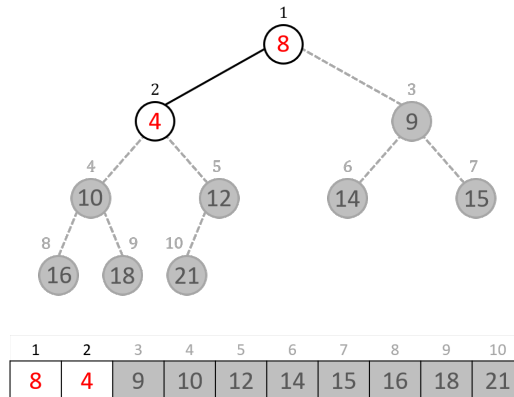
Diagram (no need to include in your answer):



2(i) [4 Points] Show state of $A[1 : 10]$ after removing the 8th largest item of A from the heap and max-heapifying.

8, 4, 9, 10, 12, 14, 15, 16, 18, 21

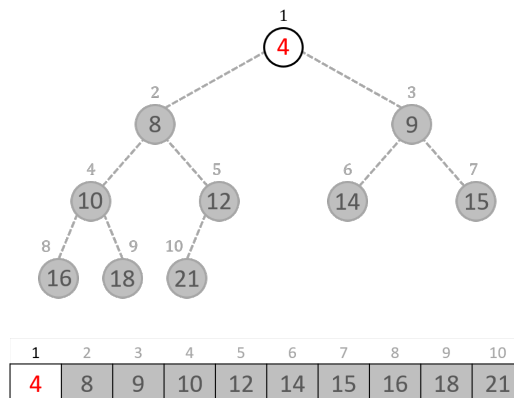
Diagram (no need to include in your answer):



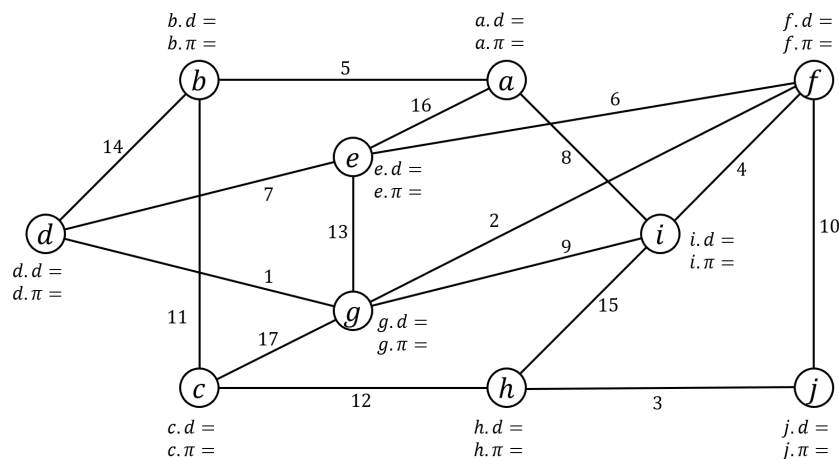
2(j) [4 Points] Show state of $A[1 : 10]$ after removing the 9th largest item of A from the heap and max-heapifying.

4, 8, 9, 10, 12, 14, 15, 16, 18, 21

Diagram (no need to include in your answer):



QUESTION 3. [65 Points] Prim's MST. Recall Prim's Minimum Spanning Tree (MST) algorithm we saw in the class. This task asks you for a step-by-step demonstration of how the algorithm find an MST of the following weighted undirected graph.



We will start growing an MST from node a . Recall that the algorithm first initializes the d and π fields of each node of the graph, and then in each step adds one node to the MST and updates the d and π fields of other nodes.

In your answers you can write “**pi**” instead of π .

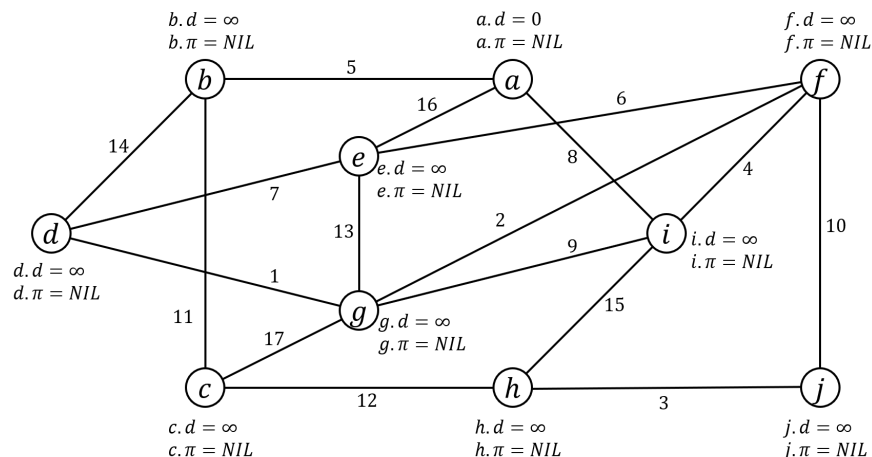
- 3(a) [5 Points] **INITIALIZATION:** What are the initial values of the d and π fields of each vertex?
- 3(b) [5 Points] **STEP 1:** Write down the 1st vertex to be added to the MST and the vertices whose d and π fields change (along with their new d and π values) as a result of relaxation.
- 3(c) [5 Points] **STEP 2:** Write down the 1st edge to be added to the MST and the vertices whose d and π fields change (along with their new d and π values).
- 3(d) [5 Points] **STEP 3:** Repeat 3(c) for the 2nd edge to be added to the MST.
- 3(e) [5 Points] **STEP 4:** Repeat 3(c) for the 3rd edge to be added to the MST.
- 3(f) [5 Points] **STEP 5:** Repeat 3(c) for the 4th edge to be added to the MST.
- 3(g) [5 Points] **STEP 6:** Repeat 3(c) for the 5th edge to be added to the MST.
- 3(h) [5 Points] **STEP 7:** Repeat 3(c) for the 6th edge to be added to the MST.
- 3(i) [5 Points] **STEP 8:** Repeat 3(c) for the 7th edge to be added to the MST.
- 3(j) [5 Points] **STEP 9:** Repeat 3(c) for the 8th edge to be added to the MST.
- 3(k) [5 Points] **STEP 10:** Repeat 3(c) for the 9th edge to be added to the MST.
- 3(l) [10 Points] What is the weight of the MST you have just computed?

SOLUTION.

3(a) [5 Points] INITIALIZATION: What are the initial values of the d and π fields of each vertex?

The (d, π) fields of vertex a are set to $(0, \text{NIL})$, and those of all other vertices are set to $(\text{infinity}, \text{NIL})$.

Diagram (no need to include in your answer):

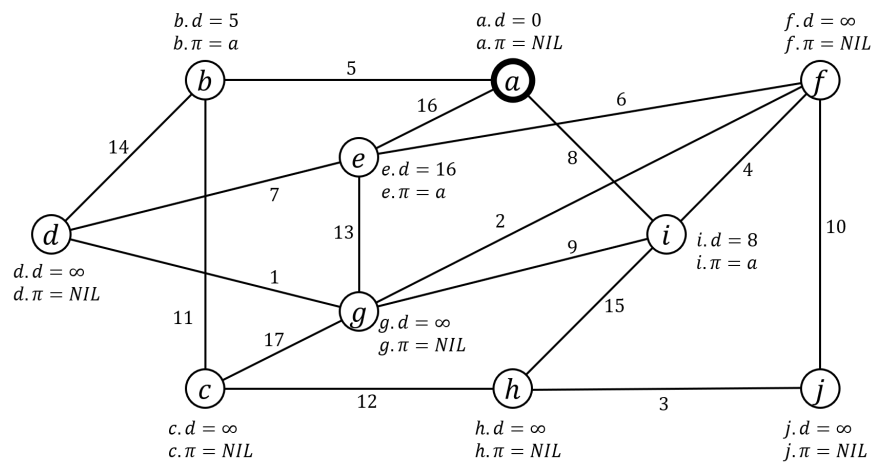


3(b) [5 Points] STEP 1: Write down the 1st vertex to be finalized and the vertices whose d and π fields change (along with their new d and π values) as a result of relaxation.

Vertex added: a

Changed (d, π) : $b(5, a)$, $e(16, a)$, $i(8, a)$

Diagram (no need to include in your answer):

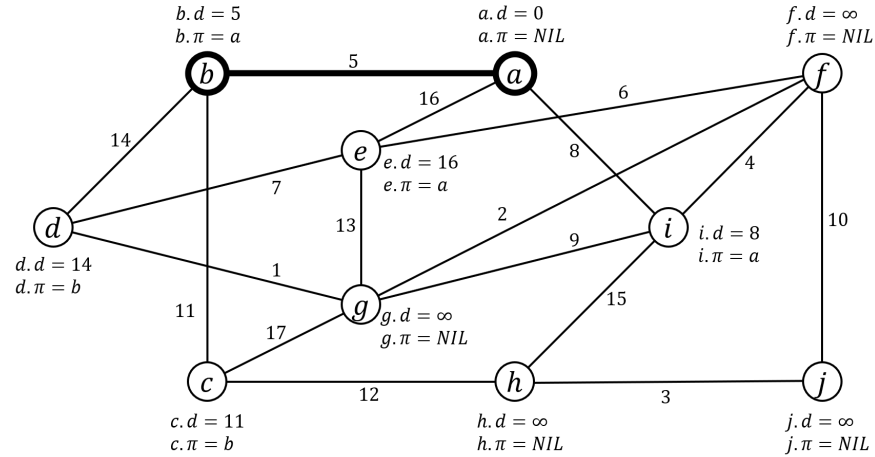


3(c) [5 Points] STEP 2: Write down the 1st edge to be added to the MST and the vertices whose d and π fields change (along with their new d and π values).

Edge added: (a, b)

Changed (d, π) : c (11, b), d (14, b)

Diagram (no need to include in your answer):

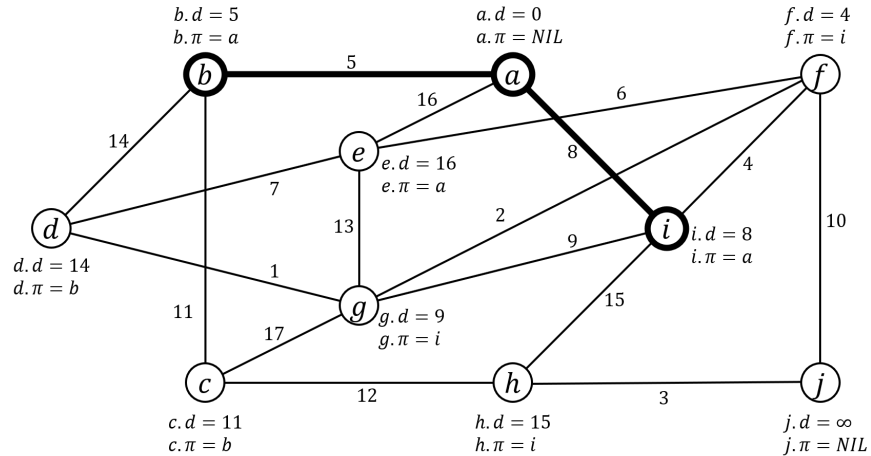


3(d) [5 Points] STEP 3: Repeat 3(c) for the 2nd edge to be added to the MST.

Edge added: (a, i)

Changed (d, π) : f (4, i), g (9, i), h (15, i)

Diagram (no need to include in your answer):

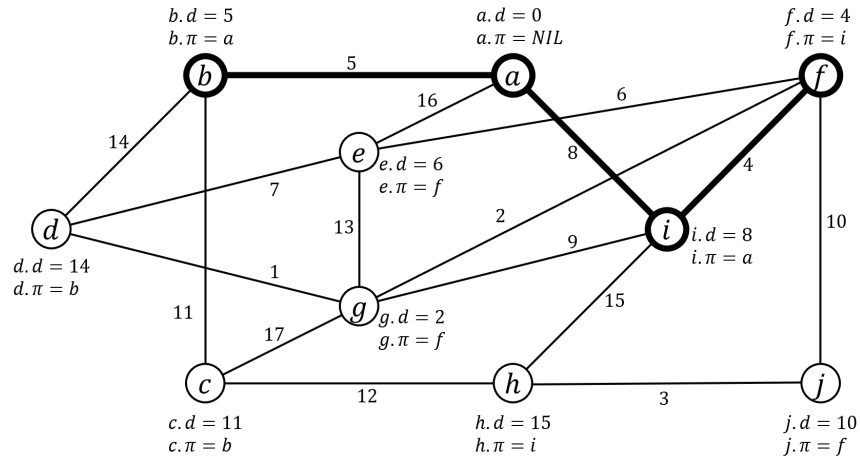


3(e) [**5 Points**] STEP 4: Repeat 3(c) for the 3rd edge to be added to the MST.

Edge added: (i, f)

Changed (d, π) : $e(6, f)$, $g(2, f)$, $j(10, f)$

Diagram (no need to include in your answer):

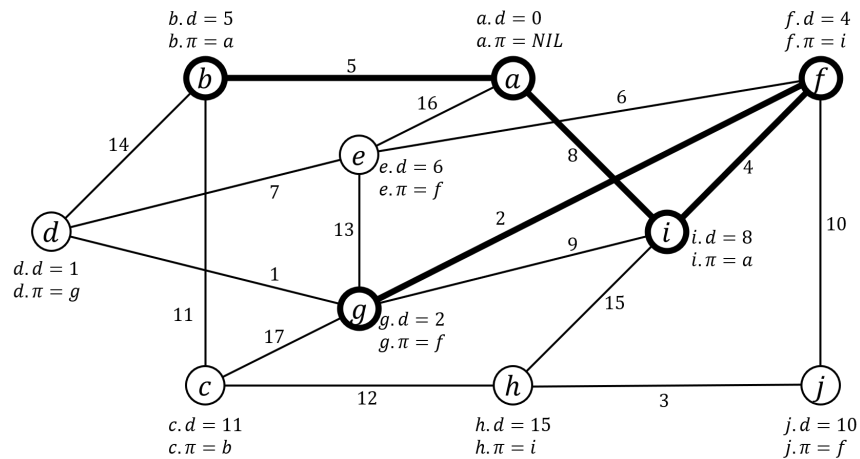


3(f) [**5 Points**] STEP 5: Repeat 3(c) for the 4th edge to be added to the MST.

Edge added: (f, g)

Changed (d, π) : $d(1, g)$

Diagram (no need to include in your answer):

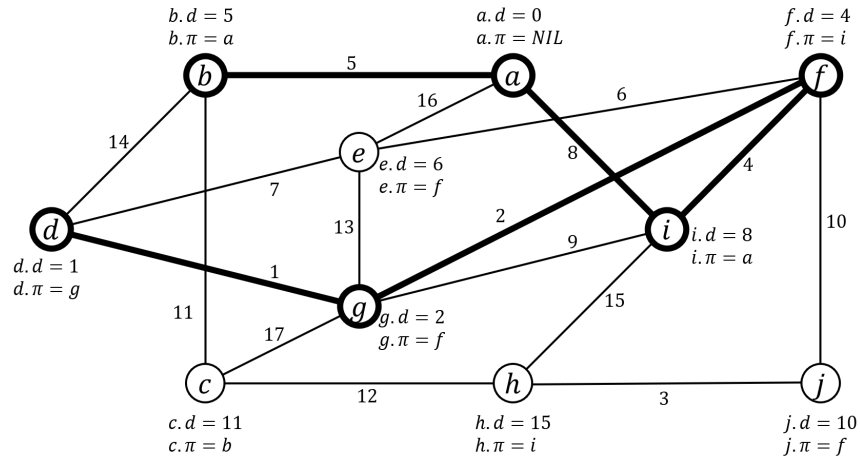


3(g) [5 Points] STEP 6: Repeat 3(c) for the 5th edge to be added to the MST.

Edge added: (g, d)

Changed (d, π) : none

Diagram (no need to include in your answer):

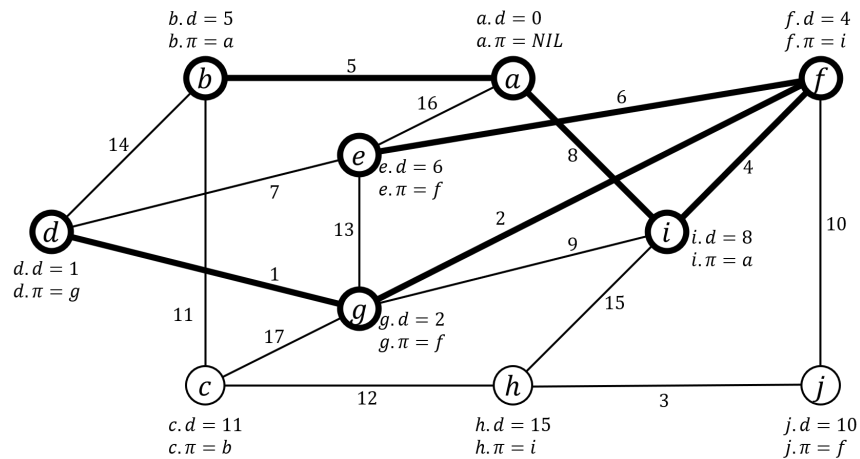


3(h) [5 Points] STEP 7: Repeat 3(c) for the 6th edge to be added to the MST.

Edge added: (f, e)

Changed (d, π) : none

Diagram (no need to include in your answer):

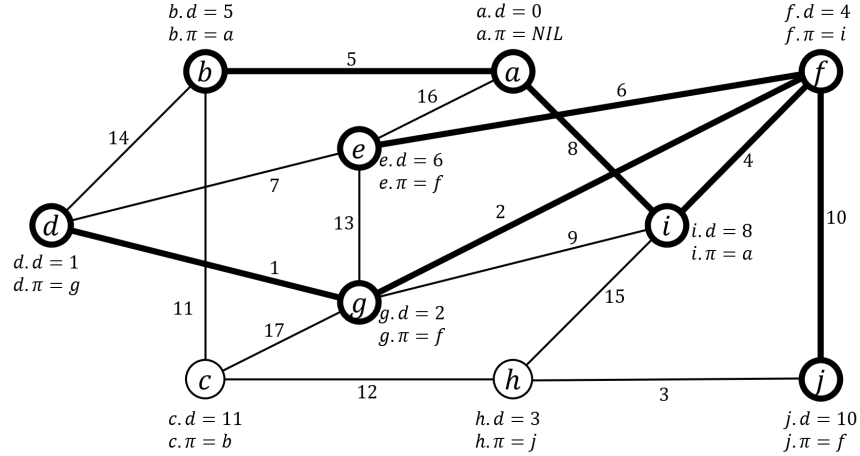


3(i) [5 Points] STEP 8: Repeat 3(c) for the 7th edge to be added to the MST.

Edge added: (f, j)

Changed (d, π) : $h(3, j)$

Diagram (no need to include in your answer):

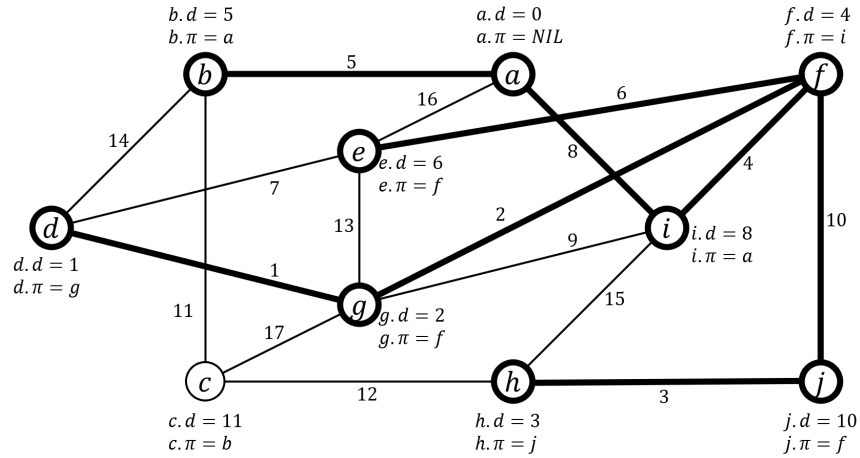


3(j) [5 Points] STEP 9: Repeat 3(c) for the 8th edge to be added to the MST.

Edge added: (j, h)

Changed (d, π) : none

Diagram (no need to include in your answer):

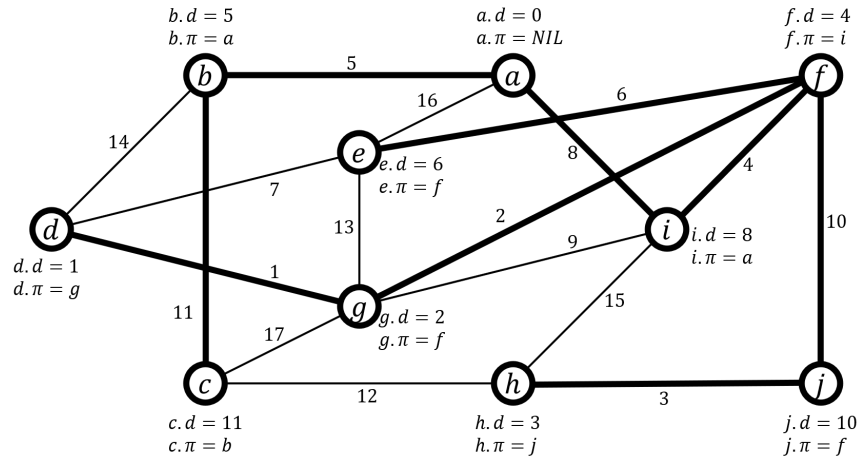


3(k) [**5 Points**] STEP 10: Repeat 3(c) for the 9th edge to be added to the MST.

Edge added: (b, c)

Changed (d, pi) : none

Diagram (no need to include in your answer):



3(l) [**10 Points**] What is the weight of the MST you have just computed?

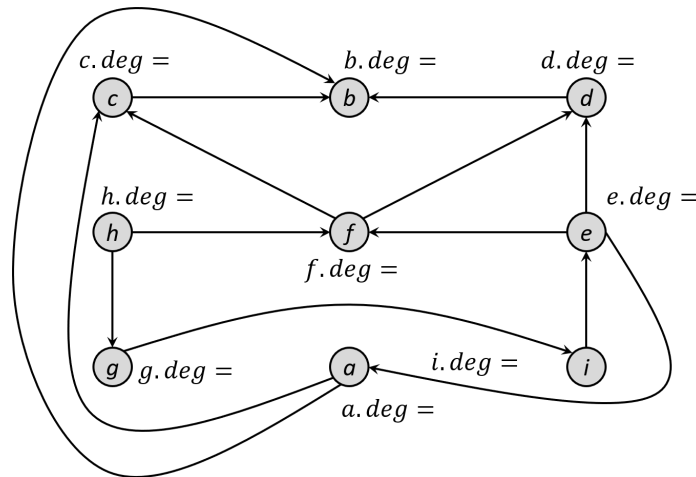
Total weight of the MST: $w(a, b) + w(a, i) + w(i, f) + w(f, g) + w(g, d) + w(f, e) + w(f, j) + w(j, h) + w(b, c) = 5 + 8 + 4 + 2 + 1 + 6 + 10 + 3 + 11 = 50$.

QUESTION 4. [55 Points] Topological Sort. Recall that a topological sort of a DAG (i.e., directed acyclic graph) $G = (V, E)$ is a linear ordering of all its vertices such that if G contains an edge (u, v) , then u appears before v in the ordering.

One can view a topological sort of a graph as an ordering of its vertices along a horizontal line so that all directed edges go from left to right.

The topological sort algorithm we saw in the class removes the vertices from the DAG and puts them on that horizontal line one by one from left to right based on vertex degrees. When a vertex v is removed from the DAG we say that it is finalized.

In this task you are given the following DAG and you are required to find in which order the vertices will be finalized and how the vertex in-degrees will be changed as a result.



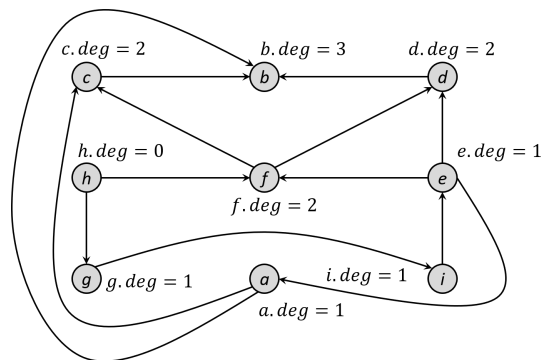
- 4(a) [5 Points] Write down the in-degree of each vertex in the DAG.
- 4(b) [5 Points] Write down the 1st vertex to be finalized and list the vertices whose in-degrees change (along with their updated in-degrees) as a result.
- 4(c) [5 Points] Repeat 4(b) for the 2nd vertex to be finalized.
- 4(d) [5 Points] Repeat 4(b) for the 3rd vertex to be finalized.
- 4(e) [5 Points] Repeat 4(b) for the 4th vertex to be finalized.
- 4(f) [5 Points] Repeat 4(b) for the 5th vertex to be finalized.
- 4(g) [5 Points] Repeat 4(b) for the 6th vertex to be finalized.
- 4(h) [5 Points] Repeat 4(b) for the 7th vertex to be finalized.
- 4(i) [5 Points] Repeat 4(b) for the 8th vertex to be finalized.
- 4(j) [5 Points] Repeat 4(b) for the 9th vertex to be finalized.
- 4(k) [5 Points] Write down the vertices in topologically sorted order.

SOLUTION.

4(a) [5 Points] Write down the in-degree of each vertex in the DAG.

$a.deg = 1, b.deg = 3, c.deg = 2,$
 $d.deg = 2, e.deg = 1, f.deg = 2,$
 $g.deg = 1, h.deg = 0, i.deg = 1$

Diagram (no need to include in your answer):

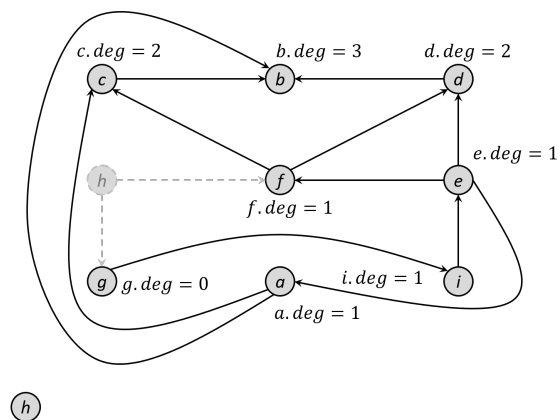


4(b) [5 Points] Write down the 1st vertex to be finalized and list the vertices whose in-degrees change (along with their updated in-degrees) as a result.

Finalized: h

In-degree changed: g ($g.deg = 0$)

Diagram (no need to include in your answer):

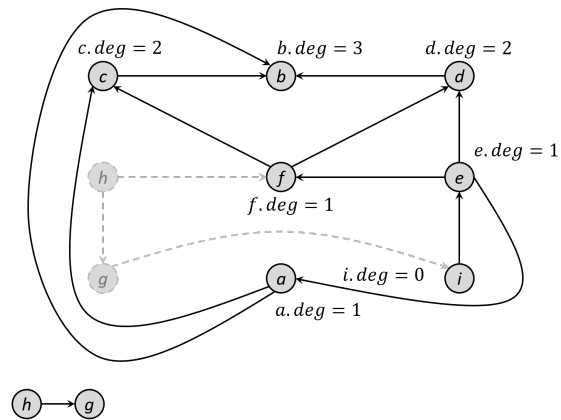


4(c) [**5 Points**] Repeat 4(b) for the 2nd vertex to be finalized.

Finalized: g

In-degree changed: i ($i.deg = 0$)

Diagram (no need to include in your answer):

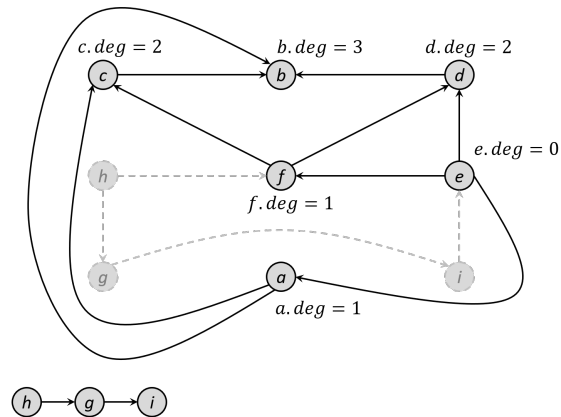


4(d) [**5 Points**] Repeat 4(b) for the 3rd vertex to be finalized.

Finalized: i

In-degree changed: e ($e.deg = 0$)

Diagram (no need to include in your answer):

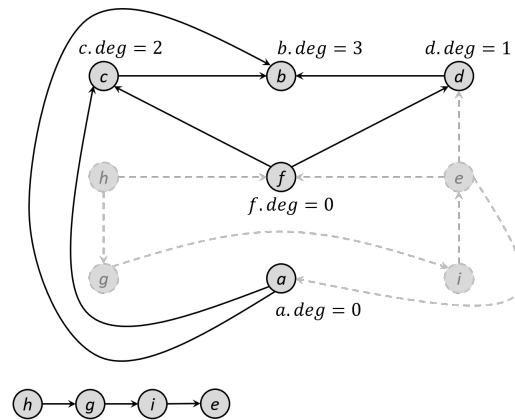


4(e) [**5 Points**] Repeat 4(b) for the 4th vertex to be finalized.

Finalized: e

In-degree changed: a ($a.deg = 0$), d ($d.deg = 1$), f ($f.deg = 0$)

Diagram (no need to include in your answer):

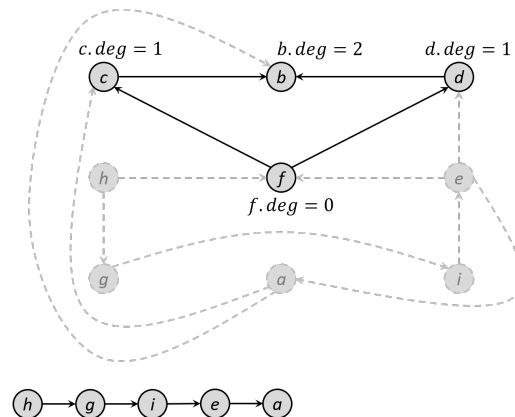


4(f) [**5 Points**] Repeat 4(b) for the 5th vertex to be finalized.

Finalized: a

In-degree changed: b ($b.deg = 2$), c ($c.deg = 1$)

Diagram (no need to include in your answer):

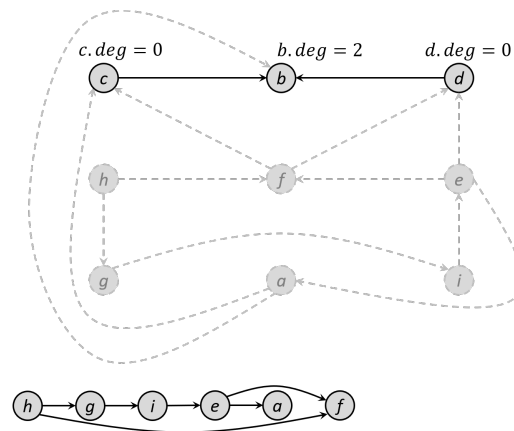


4(g) [**5 Points**] Repeat 4(b) for the 6th vertex to be finalized.

Finalized: f

In-degree changed: c ($c.deg = 0$), d ($d.deg = 0$)

Diagram (no need to include in your answer):

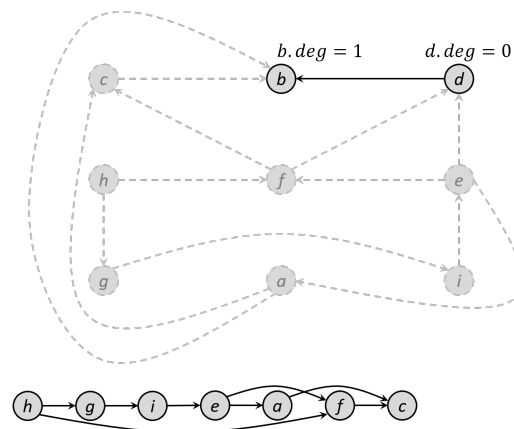


4(h) [**5 Points**] Repeat 4(b) for the 7th vertex to be finalized.

Finalized: c

In-degree changed: b ($b.deg = 1$)

Diagram (no need to include in your answer):

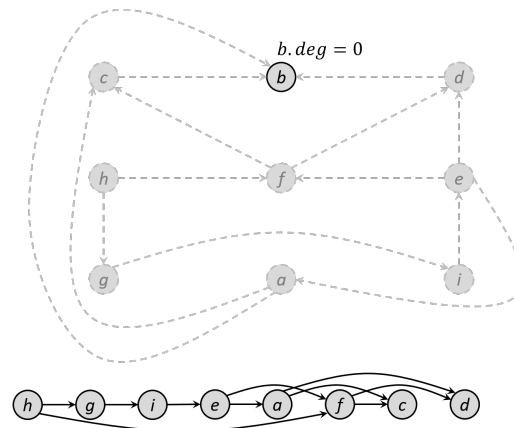


4(i) [5 Points] Repeat 4(b) for the 8th vertex to be finalized.

Finalized: d

In-degree changed: b ($b.deg = 0$)

Diagram (no need to include in your answer):

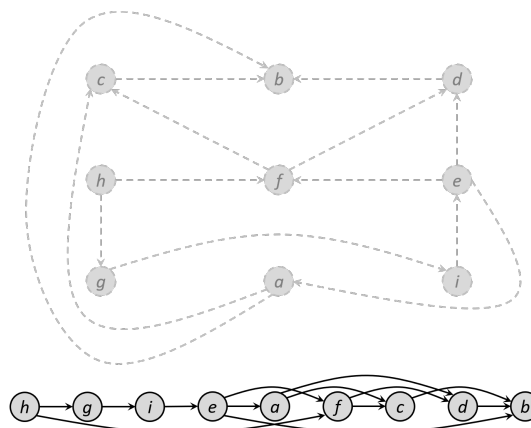


4(j) [5 Points] Repeat 4(b) for the 9th vertex to be finalized.

Finalized: b

In-degree changed: none

Diagram (no need to include in your answer):



4(k) [5 Points] Write down the vertices in topologically sorted order.

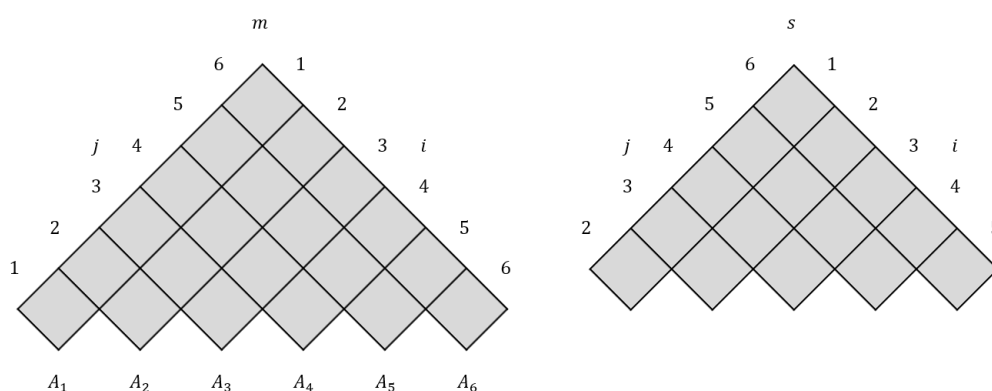
$h, g, i, e, a, f, c, d, b$

QUESTION 5. [40 Points] Matrix Chain Multiplication. Recall that the matrix-chain multiplication problem is as follows: given a chain $\langle A_1, A_2, \dots, A_n \rangle$ of n matrices, where for $i = 1, 2, \dots, n$ matrix A_i has dimension $p_{i-1} \times p_i$, fully parenthesize the product $A_1 \times A_2 \times \dots \times A_n$ in a way that minimizes the number of scalar multiplications.

In this task, you are given the following chain of six (6) matrices.

matrix	A_1	A_2	A_3	A_4	A_5	A_6
dimension	6×3	3×7	7×3	3×1	1×2	2×5

Your task is to fill out the following two tables the way we saw in the class.



5(a) [20 Points] Fill out the tables given above.

For the m table: for each $k \in [1, 6]$, write down all entries $m[i, j]$ with $j - i = 6 - k$ from on line k . Put the entries on the same line in increasing order of i values.

For the s table: for each $k \in [1, 5]$, write down all entries $s[i, j]$ with $j - i = 6 - k$ from on line k . Put the entries on the same line in increasing order of i values.

5(b) [8 Points] What is the smallest number of scalar multiplications needed to multiply the six input matrices? Which entry of which table stores this value?

5(c) [12 Points] Write down the parenthesization of $A_1 \times A_2 \times \dots \times A_n$ that minimizes the number of scalar multiplications needed to multiply them. Which table do you extract it from?

SOLUTION.

5(a) [**20 Points**] Fill out the tables given above.

For the m table: for each $k \in [1, 6]$, write down all entries $m[i, j]$ with $j - i = 6 - k$ from on line k . Put the entries on the same line in increasing order of i values.

For the s table: for each $k \in [1, 5]$, write down all entries $s[i, j]$ with $j - i = 6 - k$ from on line k . Put the entries on the same line in increasing order of i values.

Table m :

100

72, 67

60, 48, 66

117, 42, 35, 25

126, 63, 21, 6, 10

0, 0, 0, 0, 0, 0

Table s :

4

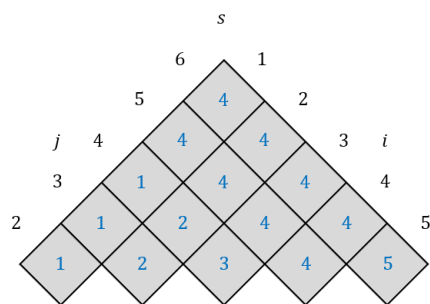
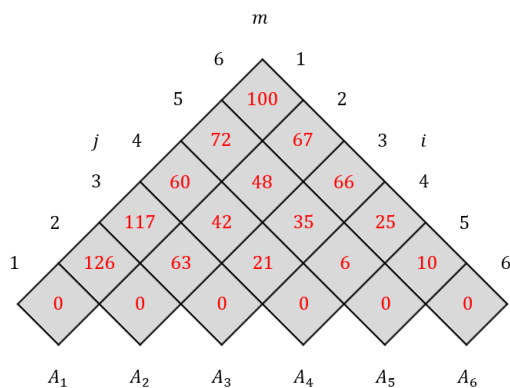
4, 4

1, 4, 4

1, 2, 4, 4

1, 2, 3, 4, 5

Diagram (no need to include in your answer):



- 5(b) [**10 Points**] What is the smallest number of scalar multiplications needed to multiply the six input matrices? Which entry of which table stores this value?

The smallest number of scalar multiplications need: 100.

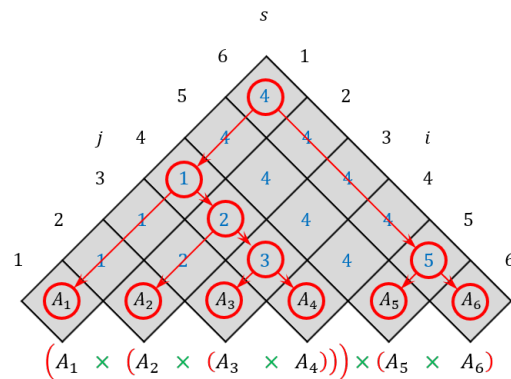
This number is stored in $m[1, 6]$.

- 5(c) [**10 Points**] Write down the parenthesization of $A_1 \times A_2 \times \dots \times A_n$ that minimizes the number of scalar multiplications needed to multiply them. Which table do you extract it from?

Here is the optimal parenthesization: $(A_1 \times (A_2 \times (A_3 \times A_4))) \times (A_5 \times A_6)$.

The ordering above is obtained from the s table.

Diagram (no need to include in your answer):



QUESTION 6. [30 Points] Complexities. For each of the problems given below, write down which problem we saw in the class it corresponds to. If there is a polynomial-time algorithm for solving the problem, state the running time of that algorithm. If the problem is NP-Complete, NP-Hard, or has no known polynomial time algorithm just write “Hard” instead of stating the running time.

No justifications necessary, but you must state what each variable in your stated complexity means (e.g., meanings of m and n in $\mathcal{O}((m+n)\log\log m)$).

- 6(a) [7.5 Points] Suppose you want to visit all major airports in the US. Not every airport has a direct flight to every other airport. You want to determine if it is possible to fly through every airport exactly once.
- 6(b) [7.5 Points] Stony Brook University has a number of buildings that they want to connect with tunnels or covered bridges so that there exists a way to walk from any building to any other building without going outside. In the interest of saving money, SBU wants the total length of the bridges to be as small as possible.
- 6(c) [7.5 Points] Suppose a museum has many galleries and they are connected by corridors. Every gallery is connected to one or more other galleries in that way. If a guard is installed in a gallery he/she can keep an eye on every corridor connected to that gallery. The museum authorities want to place the minimum number of guards in the galleries so that together they can guard all corridors.
- 6(d) [7.5 Points] Suppose you need to take a specific set of courses during your undergraduate years. But each course has zero or more other courses as prerequisites. In which order you should take the courses so that all prerequisites are satisfied?

SOLUTION.

- 6(a) [**7.5 Points**] Suppose you want to visit all major airports in the US. Not every airport has a direct flight to every other airport. You want to determine if it is possible to fly through every airport exactly once.

Corresponds to the Hamiltonian Cycle problem which is **Hard**.

- 6(b) [**7.5 Points**] Stony Brook University has a number of buildings that they want to connect with tunnels or covered bridges so that there exists a way to walk from any building to any other building without going outside. In the interest of saving money, SBU wants the total length of the bridges to be as small as possible.

Corresponds to the Minimum Spanning Tree problem which can be solved in $\mathcal{O}((m + n) \log n)$, where n is the number of vertices (i.e., number of buildings) and m is the number of edges (i.e., number of tunnels or covered bridges) in the input graph.

[Note: $\mathcal{O}(m + n \log n)$ is also acceptable which is obtained using a Fibonacci heap.]

- 6(c) [**7.5 Points**] Suppose a museum has many galleries and they are connected by corridors. Every gallery is connected to one or more other galleries in that way. If a guard is installed in a gallery he/she can keep an eye on every corridor connected to that gallery. The museum authorities want to place the minimum number of guards in the galleries so that together they can guard all corridors.

Corresponds to the Vertex Cover problem which is **Hard**.

- 6(d) [**7.5 Points**] Suppose you need to take a specific set of courses during your undergraduate years. But each course has zero or more other courses as prerequisites. In which order you should take the courses so that all prerequisites are satisfied?

Corresponds to the Topological Sort problem which can be solved in $\mathcal{O}((m + n))$, where n is the number of vertices (i.e., number of courses) and m is the number of edges (i.e., total number of prerequisite relationships) in the input DAG..